SYLLABUS : ELASTICITY & VISCOSITY + SURFACE TENSION

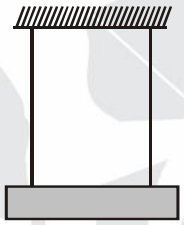
1. The diameter of a brass rod is 4 mm and Young's modulus of brass is 9×10^{10} N/m^2. The force required to stretch it by 0.1% of its length is :

(A) $360\,\pi$ N (B) 36 N (C) $144\,\pi \times 10^3$ N (D) $36\,\pi \times 10^5$ N

2. A steel wire is suspended vertically from a rigid support. When loaded with a weight in air, it expands by L_a and when the weight is immersed completely in water, the extension is reduced to L_w. Then relative density of the material of the weight is

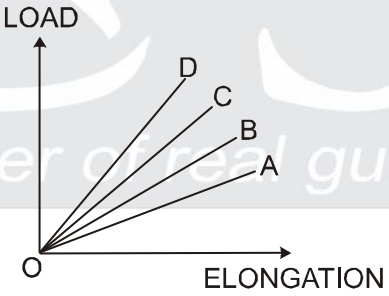
(A) $\frac{L_a}{L_a - L_w}$ (B) $\frac{L_w}{L_a}$ (C) $\frac{L_a}{L_w}$ (D) $\frac{L_w}{L_a - L_w}$

3. Two wires of equal length and cross-section area suspended as shown in figure. Their Young's modulus are Y_1 and Y_2 respectively. The equivalent Young's modulus will be



(A) $Y_1 + Y_2$ (B) $\frac{Y_1 + Y_2}{2}$ (C) $\frac{Y_1 Y_2}{Y_1 + Y_2}$ (D) $\sqrt{Y_1 \quad Y_2}$

4. The load versus elongation graph for four wires of the same materials is shown in the figure. The thinnest wire is represented by the line :



(A) OC (B) OD (C) OA (D) OB

5. A square brass plate of side 1.0 m and thickness 0.005 m is subjected to a force F on its smaller opposite edges, causing a displacement of 0.02 cm. If the shear modulus of brass is 0.4×10^{11} N/m^2, the value of the force F is

(A) 4×10^3 N (B) 400 N (C) 4×10^4 N (D) 1000 N

6. A metal block is experiencing an atmospheric pressure of 1×10^5 N/m^2, when the same block is placed in a vacuum chamber, the fractional change in its volume is (the bulk modulus of metal is 1.25×10^{11} N/m^2)

(A) 4×10^{-7} (B) 2×10^{-7} (C) 8×10^{-7} (D) 1×10^{-7}

7. A steel rod of length ℓ area of cross section A, Young's modulus Y and linear coefficient of expansion α is heated through T°C. The work that can be performed by the rod when heated is

(A) $(YA\alpha t) \times (\ell\alpha t)$ (B) $\frac{1}{2}(YA\alpha t)\times(\ell\alpha t)$ (C) $\frac{1}{2}(YA\alpha t)\times\frac{1}{2}(\ell\alpha t)$ (D) $2(YA\alpha t)\times(\ell\alpha t)$

8. A uniform wire of cross sectional area A and Young's modulus Y is stretched within elastic limit. If S is the stress in the wire, the elastic energy density stored in the wire in terms of the given parameters is

(A) $\frac{S}{2Y}$ (B) $\frac{2Y}{S^2}$ (C) $\frac{S^2}{2Y}$ (D) $\frac{S^2}{Y}$

9. If the potential energy of a spring is V on stretching it by 2 cm, then its potential energy when it is stretched by 10 cm will be :

(A) V/25 (B) 5 V (C) V/5 (D) 25 V

10. If work done in stretching a wire by 1mm is 2J, the work necessary for stretching another wire of same material, but with double the radius and half the length by 1mm in joule is -

(A) 1/4 (B) 4 (C) 8 (D) 16

11. A metal wire of length L area of cross-section A and Young's modulus Y is stretched by a variable force F such that F is always slightly greater than the elastic force of resistance in the wire. When the elongation of the wire is ℓ :

(A) the work done by F is $\frac{YA^2}{L}$ (B) the work done by F is $\frac{YA\ell^2}{2L}$

(C) heat is produced during the elongation (D) None of these

12. The terminal velocity of a sphere moving through a viscous medium is :

(A) directly proportional to the radius of the sphere

(B) inversely proportional to the radius of the sphere

(C) directly proportional to the square of the radius of sphere

(D) inversely proportional to the square of the radius of sphere

13. A sphere is dropped gently into a medium of infinite extent. As the sphere falls, the force acting downwards on it

(A) remains constant throughout

(B) increases for sometime and then becomes constant

(C) decreases for sometime and then becomes zero

(D) increases for sometime and then decreases.

14. A solid sphere falls with a terminal velocity of 10 m/s in air. If it is allowed to fall in vacuum,

(A) terminal velocity will be more than 10 m/s

(B) terminal velocity will be less than 10 m/s

(C) terminal velocity will be 10 m/s

(D) there will be no terminal velocity

15. Spherical balls of radius R are falling in a viscous fluid of viscosity η with a velocity v. The retarding viscous force acting on the spherical ball is :

(A) directly proportional to R but inversely proportional to v

(B) directly proportional to both radius R and velocity v

(C) inversely proportional to both radius R and velocity v

(D) inversely proportional to R but directly proportional to v

16. A force F is needed to break a copper wire having radius R. The force needed to break a copper wire of radius 2 R will be :

(A) F/2 (B) 2 F (C) 4 F (D) F/4

17. Two hail stones with radii in the ratio of 1 : 2 fall from a great height through the atmosphere. Then the ratio of their momentum after they have attained terminal velocity is

(A) 1 : 1 (B) 1 : 4 (C) 1 : 16 (D) 1 : 32

18. A 50 kg motor rests on four cylindrical rubber blocks. Each block has a height of 4 cm and a cross-sectional area of 16 cm^2. The shear modulus of rubber is 2×10^6 N/m^2. A sideways force of 500 N is applied to the motor. The distance that the motor moves sideways is

(A) 0.156 cm (B) 1.56 cm (C) 0.312 cm (D) 0.204 cm

19. A small steel ball falls through a syrup at constant speed of 10 cm/s. If the steel ball is pulled upwards with a force equal to twice its effective weight, how fast will it move upwards ?

(A) 10 cm/s (B) 20 cm/s (C) 5 cm/s (D) – 5 cm/s

20. Two identical thin metal strips, one of aluminum and the other of iron are riveted together to form a bimetallic strip. The temperature is raised by 50° C. If the central planes of the two strips are separated by 2nm and the coefficients of thermal expansion of aluminum and iron are respectively $30 \times 10^{-6}/^0$ C and 10×10^{-6} $/^0$ C the average radius of curvature of the bimetallic strip is about.

(A) 50 cm (B) 100 cm (C) 150 cm (D) 200 cm

21. The pressure that has to be applied to the ends of a steel wire of length 10 cm to keep its length constant when its temperature is raised by 100°C is :

(For steel Young's modulus is 2×10^{11} N m^{-2} and coefficient of thermal expansion is 1.1×10^{-5} K^{-1})

(A) 2.2×10^8 Pa (B) 2.2×10^9 Pa (C) 2.2×10^7 Pa (D) 2.2×10^6 Pa

22. A pendulum made of a uniform wire of cross sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to T_M. If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to : (g = gravitational acceleration)

(A) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right]\frac{A}{Mg}$ (B) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right]\frac{Mg}{A}$ (C) $\left[1 - \left(\frac{T_M}{T}\right)^2\right]\frac{A}{Mg}$ (D) $\left[1 - \left(\frac{T}{T_M}\right)^2\right]\frac{A}{Mg}$

23. A rod, of length L at room temperature and uniform area of cross section A, is made a metal having coefficient of linear expansion α/°C. It is observed that an external compressive force F, is applied on each of its ends, prevents any change in the length of the rod, when its temperature rises by ΔT K. Young's modulus , Y, for this metal is :

(A) $\frac{2F}{A\alpha\Delta T}$ (B) $\frac{F}{A\alpha(\Delta T - 273)}$ (C) $\frac{F}{A\alpha\Delta T}$ (D) $\frac{F}{2A\alpha\Delta T}$

24. A load of mass M kg is suspended from a steel wire of length 2 m and radius 1.0 mm in Searle's apparatus experiment. The increase in length produced in the wire is 4.0 mm. Now the load is fully immersed in a liquid of relative density 2. The relative density of the material of load is 8. The new value of increase in length of the steel wire is:

(A) 4.0 mm (B) zero (C) 5.0 mm (D) 3.0 mm

25. In an experiment to verify Stokes law, a small spherical ball of radius r and density ρ falls under gravity through a distance h in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of h is proportional to (ignore viscosity of air)

(A) r^4 (B) r^3 (C) r (D) r^2

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ANSWER KEY

1.	(A)	**2.**	(A)	**3.**	(B)	**4.**	(C)	**5.**	(C)
6.	(C)	**7.**	(B)	**8.**	(C)	**9.**	(D)	**10.**	(D)
11.	(B)	**12.**	(C)	**13.**	(C)	**14.**	(D)	**15.**	(B)
16.	(C)	**17.**	(D)	**18.**	(A)	**19.**	(A)	**20.**	(D)
21.	(A)	**22.**	(A)	**23.**	(C)	**24.**	(D)	**25.**	(A)